[I225] Statistical Signal Processing(E) Office Hour 5

1. Suppose the following Bernoulli observations are obtained:

$$\mathbf{x} = [1,0,1,1,0].$$

Each observation follows

$$X_i \sim \text{Bernoulli}(p), \quad i = 1,2,\ldots,N,$$

where the parameter represents the probability of success.

(a) Derive the likelihood function.

(b) Derive the log-likelihood function.

(c) Find the maximum likelihood estimator.

(d) Explain why log-likelihood is commonly used.

2. Suppose the observed random vector is

$$\mathbf{X} = [X_0, X_1, \ldots, X_{N-1}]^T,$$

where each sample follows

$$X_n \sim \mathcal{N}(\mu, \sigma^2), \quad n = 0,1,\ldots,N-1.$$

Assume that the variance is known.

(a) Derive the likelihood function.

(b) Derive the log-likelihood function.

(c) Find the maximum likelihood estimator of the mean.

(d) Explain the meaning of the obtained estimator.

3. Suppose the random variable follows a Poisson distribution:

$$X \sim \text{Poisson}(\lambda),$$

with probability mass function

$$p(x;\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}.$$

Assume that independent observations are collected.

(a) Verify the regularity condition.

(b) Compute the Fisher information.

(c) Find the Cramer-Rao lower bound.

(d) Explain the relationship between Fisher information and estimation accuracy.

4. An engineer is tracking a drone that is accelerating away from a base station. The position $x[n]$ at discrete time step n is modeled by a quadratic polynomial with three unknown parameters: initial position (θ_1), initial velocity (θ_2), and constant acceleration (θ_3):

$$\hat{s}[n] = \theta_1 + \theta_2 n + \theta_3 n^2$$

We collected 4 noisy measurements at times $n = 0,1,2,3$: $\boldsymbol{s} = [2,6,13,26]^T$.

Compute the Least Square Estimation (LSE) of $\widehat{\boldsymbol{\theta}} = \left[\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3\right]^T$ and the final scalar value of the minimum cost function $J(\widehat{\boldsymbol{\theta}})$!

5. An engineer models the trend of a signal over a short duration. Initially, they assume the signal can be modeled as a simple linear ramp passing through the origin (Order 1 model):

$$\hat{s}[n] = \theta_1 n$$

Three sequential measurements are collected at times n = {0,1,2}: s[n] = {1,3,5}

After applying LSE, the coefficient is estimated as $\theta_1 = 2.6$

Later, to improve model accuracy, the engineer decides to introduce a constant intercept (DC offset) term θ_2 to create an updated, more expressive model (Order 2 model):

$$\hat{s}[n] = \theta_1 n + \theta_2$$

Instead of performing a full batch matrix inversion from scratch for the Order 2 model, use **Order-Recursive Least Squares** to gracefully update the parameters.